

EVA DURÁN CAMACHO

Astrophysics Researcher

ExGal-Twin Postdoctoral Fellow
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Employment

- Jan 2025– **Postdoctoral Fellow**, *Instituto de Astrofísica de Canarias (IAC)*, Tenerife, Spain.
Current: Numerical simulations of Milky Way type Galaxies as part of the WIDERA ExGal-Twin project funded by the European Union. Code development to improve galaxy modelling, focusing on bar formation, feedback effects, dynamics and environmental effects on star formation.
PI: Prof. Johan Knapen
- Jul–Aug **Research Assistant**, *University of Cambridge*, Cambridge, United Kingdom.
2020: Effects of magnetic fields and rotation on the evolution of the early type of stars T Tauri using the STARS code. Advisor: Prof. Christopher Tout
- Jan–Jun **Laboratory Technician**, *Universidad Autónoma*, Madrid, Spain.
2018: Extracting graphene samples from graphite at the Condensed Matter Department

Education

- 2020–24 : **PhD, Astrophysics**, *Cardiff University*, Cardiff, United Kingdom.
Supervisors: Dr. Ana Duarte Cabral & Dr. Paul Clark
- 2019–20 : **MAsT in Astrophysics**, *University of Cambridge*, Cambridge, United Kingdom.
Supervisors: Dr. Giorgia Busso, Dr. Francesca D'Angeli & Prof. Mike Irwin
- 2015–19 : **Grado en Física**, *Universidad Autónoma*, Madrid, Spain.
Supervisors: Prof. Gwendolyn Meeus & Dr. Yago Ascasibar

Summer Schools

- Jul 2021 : **Interstellar Medium of Galaxies, from the Epoch of Reionization to the Milky Way**, *International Summer School*, IAU, Virtual Edition.

Grants & Awards

- Jun 2025 **Computational Grant** Project Leader. Red Española de Supercomputación. Awarded 250 CPU hours (priority) and 22 TB storage on Tirant at Universitat de València (Spain)
- Mar 2023 **Taith Mobility Grant** (1280€), Higher Education – Research Mobility in Heidelberg (Germany)
- Aug 2022 **IAU Travel Grant** (1682€), International Astronomical Union (IAU)
- Aug 2022 **KAS Travel Grant** (318€, accommodation), Korean Astronomical Society (KAS)
- Jun 2022 **Travel Grant** (1594\$), Flatiron Institute (New York, USA)
- 2015–2019 : **Distinctions**, *Physics Degree*, *Universidad Autónoma*, Madrid, Spain.
Análisis I (1st year), Electromagnetismo I (2nd year), Métodos Matemáticos Avanzados & Electrodinámica Clásica (3rd year), Física Atómica y Molecular & Física del Cosmos (4th year)

Conferences & Seminars

- Sep 2025: **Invited Lecture**, *Advanced Numerical Simulations School*, Durham, United Kingdom, Hands-on lecture: "Hydrodynamical codes: Introduction to AREPO".

Aug 2025: **Invited Seminar**, *National Cheng Kung University, NCKU*, Tainan City, Taiwan.

Aug 2025: **XXIst Star Formation in Different Environments**, *ICISE*, Quy Nhon, Vietnam, Contributed Talk.

Jul 2025: **National Astronomy Meeting**, *Revealing the Milky Way with Gaia: Focus on Galactic dynamics in the Gaia era and beyond*, Durham, United Kingdom, Contributed Talk.

Jun 2025: **EAS Annual Meeting, SS6: A panoptic view of the solar neighbourhood: new insights into the nearest ISM laboratory**, Cork, Ireland, Poster, Short Presentation.

Mar 2025: **Invited Seminar at FLAT meetings**, *Durham University*, Durham, UK, .

Feb 2025: **Invited Seminar at ISM and Galaxies meetings**, *Cardiff University*, Cardiff, UK, .

Jan 2025: **ExGal-Twin Annual Collaboration Meeting**, Tenerife, Spain, Contributed Talk .

June 2024: **OGHReS Workshop**, Bonn, Germany, Contributed Talk .

Nov 2023: **Star formation in the Milky Way and beyond**, *RAS meeting*, London, UK, Contributed Talk .

Oct 2023: **Invited Seminar at Stefanie Walch-Gassner's Group**, Cologne, Germany.

Sept 2023: **AREPO development workshop for ISM simulations**, *Manchester, UK*, Contributed Talk.

Jul 2023: **National Astronomical Meeting**, *Astrophysical Simulations: Keeping up with the Frontier*, Cardiff, Wales, Contributed Talk .

Jul 2023: **National Astronomical Meeting**, *Star formation across the Milky Way*, Cardiff, Wales, Contributed Talk.

Jun 2023: **Star at Lyon: The Physics of Star Formation**, *Lyon*, France, Poster.

Jun 2023: **Invited Seminar at the Galaxy Evolution Research Group**, *Oxford University*, Oxford, UK.

Jun 2023: **Star Formation in the era of JWST**, *Paralia Katerinis*, Greece, Poster.

Apr 2023: **Invited Seminar at the ISM Research Group**, *Heidelberg University*, Heidelberg, Germany.

Aug 2022: **XXth Star Formation in Different Environments**, *ICISE*, Quy Nhon, Vietnam, Contributed Talk.

Aug 2022: **IAU Symposium 373: Resolving the Rise and Fall of Star Formation in Galaxies**, *IAU General Assembly XXXI*, Busan, South Korea, Poster.

Jun 2022: **From Stars to Galaxies II: Connecting our understanding of star and galaxy formation**, *Chalmers*, Gothenburg, Sweden, Poster.

Jun 2022: **Computational Astrophysics in the ngVLA Era: Synergistic Simulations, Theory, and Observations**, *Flatiron Institute*, New York, USA, Poster.

Jan 2021: **UKRI STFC Introductory Course in Astronomy for New Research Students**, *Armagh Observatory and Planetarium*, Virtual Editon , Contributed Talk.

Jun 2021: **Conference for Astronomy and Physics Students (CAPS '21)**, *University of Birmingham*, Virtual Editon , Poster.

Nov 2021: **10th IRAM 30-meter School on Milimeter Astronomy**, *Institut de Radioastronomie Milimétrique*, Virtual Editon , Attendee.

Scientific Contributions

- 2025- **Board Member**, *ExGal-Twin Collaboration*, Partner institutions: Instituto de Astrofísica de Canarias (IAC), University of Groningen (RUG), Durham University (ICC) and CNRS - Observatoire Astronomique de Strasbourg (ObAS).
- Sep 2025 **Scientific Organizing Committee (SOC)**, *Advanced Numerical Simulations School*, Durham, United Kingdom.
- Jun 2025 **Session Organizer**, *National Astronomy Meeting (NAM)*, Durham, United Kingdom, Star formation across environments: From individual molecular clouds to entire galaxies.
- Jun 2025 **Scientific Organizing Committee (SOC)**, *2nd Educado Training School*, Tenerife, Spain.
- Jan 2025 **Scientific Organizing Committee (SOC)**, *1st ExGal-Twin Annual Meeting*, Tenerife, Spain.

Publications

- 2025 **Durán-Camacho. E.**, et al., *Insights into the structure and kinematics of a Milky Way-like galaxy*, MNRAS.
- Euclid Collaboration**, et al. (including **Durán-Camacho. E.**), *Euclid Preparation. The impact of nonparametric star-formation histories on spatially resolved galaxy property estimation using synthetic Euclid images*, A&A.
- Euclid Collaboration**, et al. (including **Durán-Camacho. E.**), *Euclid Quick Data Release (Q1). A first view of the star-forming main sequence in the Euclid Deep Fields*, A&A.
- Euclid Collaboration**, et al. (including **Durán-Camacho. E.**), *Euclid Quick Data Release (Q1). A first look at the fraction of bars in massive galaxies at $z < 1$* , A&A.
- 2024 **Durán-Camacho. E.**, et al., *Self-consistent modelling of the Milky Way structure using live potentials*, MNRAS.
- 2023 **Ge, Y.**, et al. (including **Durán-Camacho. E.**), *Large-scale Velocity-coherent Filaments in the SEDIGISM Survey: Association with Spiral Arms and Fraction of Dense Gas*, A&A.

Publications in preparation

Durán-Camacho. E., et al., *Tracing Galactic Ripples: Understanding Wave Phenomena in Longitude-Velocity Diagrams of Milky Way Simulations*.

Conference Proceedings

- 2022 **Durán-Camacho. E.**, Duarte-Cabral, A., *Milky Way: structure via live potentials*. International Astronomical Union Proceedings Series

Teaching & Outreach

- Sept 2025 - **Co-supervising of PhD Student**, Cardiff University, UK.
- Sept 2029 : Co-supervision of PhD student on project "Simulated Magnetic Fields in MW-like galaxies" with Dr. Ana Duarte-Cabral
- Jul 2025 - **Co-supervising of Bachelors Student**, Universidad de La Laguna, Spain.
- May 2026 : Final year undergraduate project titled "Spiral Arms in Motion: Investigating Growth and Disruption in a Simulated Milky Way" with Dr. Arianna Di Cintio
- Oct 2024 - **Planet Hoppers Educational Game Developer**, Cardiff University, UK.
- Jan 2025 : Creation of an exoplanets database for the development of a card game to bring science closer to schools
- 2020-23 : **Marker and Demonstrator**, *1st-3rd year Physics*, Cardiff University, UK.
Introduction to Astrophysics, Computational Physics, Computational Skills for Problem Solving, Atomic and Nuclear Physics, Optics, Environmental Physics
- Jul 2023: **Outreach events at the National Astronomical Meeting (NAM)**, Volunteer, Cardiff, UK.
- Jun 2023: **Astronomy on Tap**, Organising Committee, Cardiff, UK.
- 2020-21: **The Physics Mentor Project**, Mentor, Cardiff, UK.

Computational skills

Programming: AREPO (written in Fortran & C), Python, Matlab, C++

Documents: LaTeX, Adobe, Word, Excel, PowerPoint

Plotting: DataStudio, TopCat, SciDavis, CARTA, SAOImageDS9, Glue